

Shiming Yu

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Working Experience

The Hong Kong Polytechnic University

Jan 2026 – Now

Postdoctoral Fellow (Advisor: Prof. Yuanqing Zheng)

Education

The Hong Kong Polytechnic University

Aug 2022 – Jan 2026

Ph.D. in Computer Science (Advisor: Prof. Yuanqing Zheng)

University of Electronic Science and Technology of China

Aug 2018 – Jul 2022

B.E. in Communication Engineering (GPA: 3.8/4.0)

Research Interest

AIoT, Wireless Networking

Publications

First-author papers:

[**MobiSys'25**] “Are LoRa Logical Channels Really Orthogonal? Practically Orthogonalizing Massive Logical Channels”, Shiming Yu, Ziyue Zhang, Xianjin Xia, Yuanqing Zheng, Jiliang Wang, The 23rd ACM International Conference on Mobile Systems, Applications, and Services, Anaheim, CA, USA, 2025

[**MobiCom'24**] “Revolutionizing LoRa Gateway with XGate: Scalable Concurrent Transmission across Massive Logical Channels”, Shiming Yu, Xianjin Xia, Ningning Hou, Yuanqing Zheng, Tao Gu, The 30th Annual International Conference on Mobile Computing and Networking, Washington D.C., USA, 2024

[**SenSys'24**] “FDLoRa: Tackling Downlink-Uplink Asymmetry with Full-duplex LoRa Gateways”, Shiming Yu, Xianjin Xia, Ziyue Zhang, Ningning Hou, Yuanqing Zheng, The 22nd ACM Conference on Embedded Networked Sensor Systems, Hangzhou, China, 2024

[**TMC'25**] “Resolving Inter-Logical Channel Interference for Large-scale LoRa Deployments”, Shiming Yu, Ziyue Zhang, Xianjin Xia, Yuanqing Zheng, Jiliang Wang, IEEE Transactions on Mobile Computing, 2025

[**TMC'25**] “FDLoRa: Scaling Downlink Concurrent Transmissions with Full-duplex LoRa Gateways”, Shiming Yu, Xianjin Xia, Ziyue Zhang, Ningning Hou, Yuanqing Zheng, IEEE Transactions on Mobile Computing, 2025

[**TON'25**] “XGate: Scaling LoRa Communications to Massive Logical Channels”, Shiming Yu, Xianjin Xia, Ningning Hou, Yuanqing Zheng, Tao Gu, IEEE Transactions on Networking, 2025

Other papers:

[**ICNP'25**] “SlideLoRa: Reliable Channel Activity Monitoring across Massive Logical Channels in LoRa Networks”, Jiamin Jiang, Shiming Yu, Hao Wang, Haiyang Li, Yuanqing Zheng, Lu Wang, The 33rd IEEE International Conference on Network Protocols, Seoul, South Korea, 2025

[**MobiCom'25**] “From Interference Mitigation to Toleration: Pathway to Practical Spatial Reuse in LPWANs”, Ruonan Li, Ziyue Zhang, Xianjin Xia, Ningning Hou, Wenchang Chai, Shiming Yu, Yuanqing Zheng, Tao Gu, The 31st Annual International Conference on Mobile Computing and Networking, Hong Kong, China, 2025

[**SenSys'25**] “MoLoRa: Intelligent Mobile Antenna System for Enhanced LoRa Reception in Urban Environments”, Ningning Hou, Yifeng Wang, Xianjin Xia, Shiming Yu, Yuanqing Zheng, Tao Gu, The 23rd ACM Conference on Embedded Networked Sensor Systems, Irvine, CA, USA, 2025

Professional Services

Organizing Committee

2026: Publicity Co-Chair for ACM/IEEE Symposium on Edge Computing (SEC)

Technical Program Committee

2026: MobiSys Demo & AE, AutoSys Workshop @ MobiSys, HeadSys Workshop @ MobiSys

2025: MobiCom Demo & AE, MobiSys AE

2024: MobiCom AE, MobiSys AE

Journal Reviewer

2026: TOSN, TMC, IMWUT, JNCA, Computer Networks, Computer Communications

2025: TOSN, TMC, IoTJ

2024: TOSN, IoTJ

2023: TOSN, TWC, Computer Networks

Awards

Rising Star, ACM MobiSys	2026
Distinguished Reviewer Board, ACM TOSN	2026-27
Winner (Rank #1) @ Three Minute Thesis (3MT) Competition, PolyU FCMS	2025
Student Travel Grant, ACM MobiCom	2024
Best Oral Presentation Award @ CRSC, PolyU COMP	2024
Full Ph.D. Scholarship, PolyU	2022-26
Outstanding Student Scholarship (Top 15%), UESTC	2020
National Encouragement Scholarship, UESTC	2019
Outstanding Student Scholarship (Top 15%), UESTC	2019